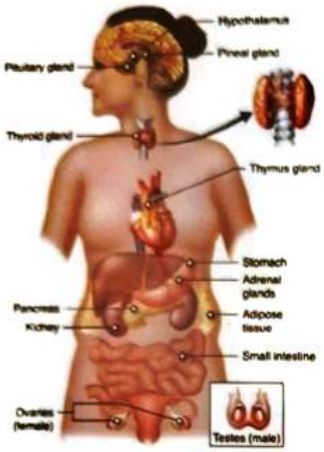
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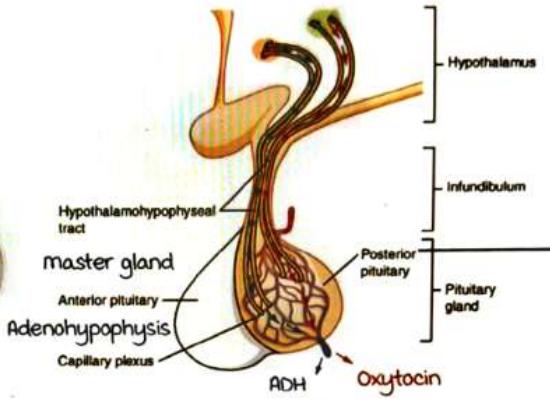
PITUITARY GLAND

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Endocrine System


Location



Parts

- No synthesis.
- Posterior pituitory also known as neurohypophysis
- Only storage (Herring bodies) & release of two hormones :
  - ADH/vasopressin.
  - Oxytocin.

Anterior Pituitary & Hypothalamus

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ANTERIOR PITUITARY :

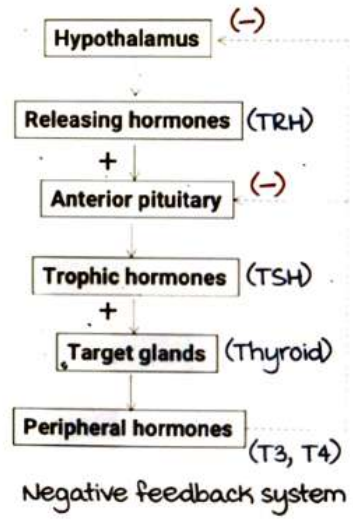
1. Acidophilic cells :

- Somatotrophs (50% of cells) : Growth Hormone
- Lactotrophs : Prolactin

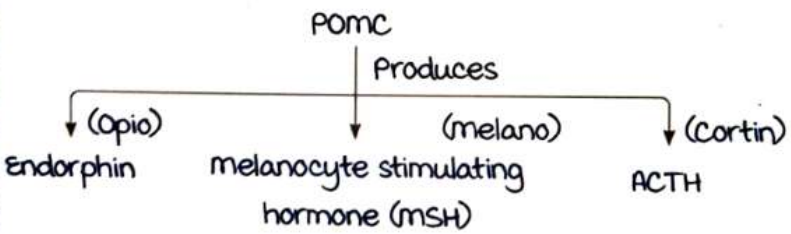
→ "Twin hormones" (Similar receptors & actions)

2. Basophilic Cells :

- Thyrotrophs : Thyroid-stimulating hormone (TSH).
- Gonadotrophs :
  - Luteinizing hormone (LH).
  - Follicle-stimulating hormone (FSH).
- Corticotrophs :
  Contain precursor molecule pro opio melanocortin (POMC).



Negative feedback system



Applied Aspect :
Addison's disease : Pathologically
↑↑↑ ACTH
↓
Action of MSH
↓
Skin hyperpigmentation

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HYPOTHALAMUS

Hypothalamic Releasing Hormones :

Hypothalamic Hormone	Hormone released from anterior pituitary
Thyrotropin - releasing hormone (TRH)	• TSH. • Prolactin.
Corticotropin - releasing hormone (CRH)	ACTH
Gonadotropin - releasing hormone (GnRH)	• LH. • FSH
Growth hormone-releasing hormone (GHRH)	GH

Hypothalamic Inhibiting Hormones :

Hypothalamic hormone -	Role in anterior pituitary
Prolactin release- inhibiting hormone (PIH)/dopamine.	Inhibits prolactin release
Growth hormone release - inhibiting hormone/somatostatin	Inhibits GH release

Anterior Pituitary Hormones

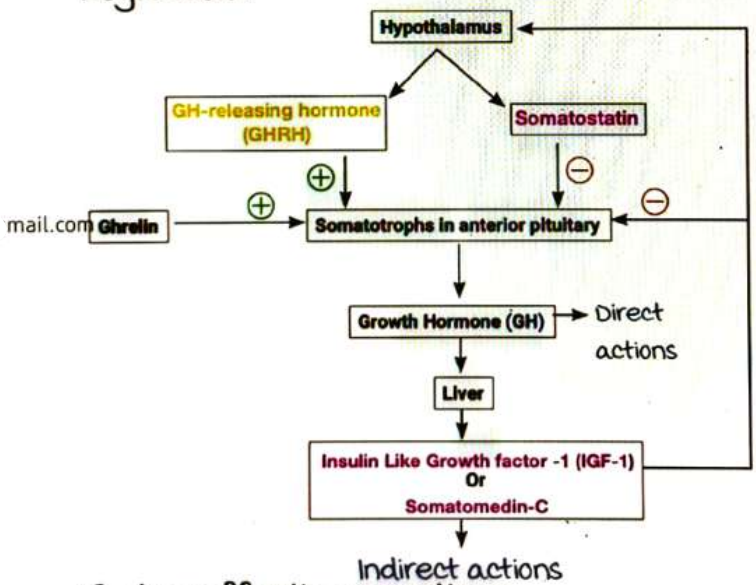
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I. GROWTH HORMONE :

Maximum levels at adolescence.

Minimum levels at senescence (Old age).

Regulation :



GHRH drug form :
- Sermorelin (↑ GH release).
- Rx of pituitary dwarfism.

Somatostatin drug form :
- Octreotide ((-) GH release).
- Rx of acromegaly.

Factors affecting secretion :

↑Growth hormone secretion	↓Growth hormone secretion
1. Hypoglycemia → GH release → ↑ Blood glucose (Diabetogenic hormone) • Fasting. • Starvation. • Exercise. • Stress. 2. Deep sleep (NREM Stage III & IV).	1. Hyperglycemia. 2. Somatostatin. 3. IGF-1. 4. REM sleep.

Feedback

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Actions :

----- Active space -----

GH —Acts on→ Cytokine receptor —Activates→ Janus Kinase

JAK - STAT pathway

Direct Actions : mediated by GH (Anti-insulin like actions).

- Lipolysis.
- Bone growth : Prechondrocytes → Chondrocytes.

Indirect Actions : mediated by IGF-1 (Insulin like actions).

- Bone Growth : All remaining actions at level of bone.

Disorders :

- Children
  - ↑GH : Gigantism
  - ↓GH : Dwarfism
- Adults
  - Acromegaly (D/t GH-releasing pituitary micro-adenoma)

Acromegaly :

Features :

- Bony overgrowth : Frontal bossing.
- ↑ hand + foot size.
- Prognathism : mandibular enlargement.
- ↑ heel pad thickness (D/t soft tissue overgrowth).
- Carpal tunnel syndrome.
- Sleep apnea (D/t upper airway obstruction).
- Diabetes mellitus.

Treatment :

- Surgical : Trans-sphenoidal resection of microadenoma
- Pharmacological
  - Somatostatin analogues : Octreotide
  - GH receptor antagonist : Pegvisomant

Laron dwarfism/GH insensitivity syndrome :

- Etiology : Defective receptors.
- Hormone levels : (N)/ ↑↑ GH, ↓ plasma levels of IGF-1.
- Rx : Mecasermin (Recombinant form of IGF-1).

Recombinant Human Growth Hormone :

Used in :

- Rx of dwarfism.
- Rx of muscle wasting in AIDS patients (↑ lean body mass & ↑ muscle mass).
- Anti-aging Rx.

Feedback

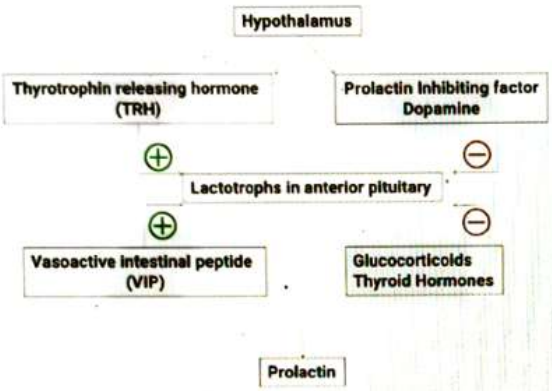
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PROLACTIN :

Stimulates milk production.

Regulation :



Applied Aspect :

Stalk effect :

Lesion at pituitary stalk ↓

Loss of releasing factors & inhibitory factors ↓

↑ Prolactin (↓ dopamine) (All other hormones ↓)

Factors affecting secretion :

↑ Secretion	↓ Secretion
• Pregnancy & lactation.	• Dopamine.
• Sexual intercourse.	• L-dopa.
• Suckling of nipples.	• Bromocriptine.
• REM sleep.	

Actions :

Prolactin —(Acts on)→ Cytokine receptor —(Activates)→ Janus kinase

JAK-STAT Pathway

- Proliferation of lobulo-alveolar epithelial cells in breast → milk production.
- Negative regulator of reproductive functions : Inhibit GnRH → Inhibit FSH & LH. ↓

  Lactational amenorrhea in ♀ (Form of contraception).

- Loss of libido & ↓ spermatogenesis in ♂.

Feedback